

--	--	--	--	--	--	--	--

B.Tech. Degree V Semester Special Supplementary Examination September 2022

ME 19-205-0505 POWER PLANT ENGINEERING (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Get a general understanding of the scale and scope of hydroelectric power stations and also grasp the basics of diesel engine and gas turbine power plants.
- CO2: Comprehend the complexity of the steam power plant after getting to know about the working of its different components and elements.
- CO3: Understand the basic principle of the working of nuclear power plant and the various types of nuclear reactors with its safety aspects, especially with respect to the current scenario of national and international debates on the same.
- CO4: Equip with the knowledge of the basics of various non-conventional energy producing methods.
- Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
- PO – Programme Outcome

PART A

(Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) List the factors which should be considered while designing a power plant.	3	3	L1	1	1
(b) Enumerate the advantages of hydroelectric power plant.	3	3	L1	1	1,2
(c) What are the different methods used for starting diesel engine power plants? Explain.	3	3	L1	1	1,2,3
(d) Write short notes on combustion chambers in gas turbines.	3	3	L1	2	1,2
(e) What is the difference between boiler mountings and accessories?	3	3	L1	3	1,2,3
(f) Explain different nuclear reactor wastes and its disposal.	3	3	L1	3	1,2,3
(g) What is solar constant and its average value? What is meant by solar insolation?	3	3	L1	4	1,2,3
(h) Write a note on OTEC.	3	3	L1	4	1,2,3

PART B

(4 × 12 = 48)

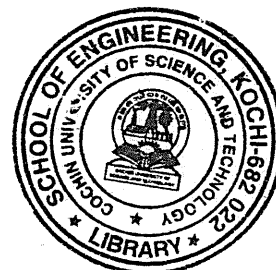
- II. (a) Which types of power plants are used as base load power plants? Why are such power plants used as base load power plants? 6 L2 1 1
- (b) A generating station has the following daily load cycle 6 L3 2 1,2

Time (Hours)	0-6	6-10	10-12	12-16	16-20	20-24
Load (MW)	40	50	60	50	70	40

Draw the load curve and find:

- maximum demand
- units generated per day
- average load
- load factor.

OR



(P.T.O)

BTS-V(SS)-09-22-1085

		Marks	BL	CO	PO
III.	(a) With a neat sketch make a note on load curves. Describe different types of load curves.	6	L2	2	1,2
	(b) A 500 kVA distribution transformer costs ₹60,000 and has a useful life of 25 years. If the salvage value is ₹5,000 and annual compound interest rate is 7.5%, determine the value of the transformer at the end of 10 years using: (i) straight line method (ii) sinking fund method	6	L3	2	1,2
IV.	Explain different methods for improving thermal efficiency of open cycle gas turbine power plant with neat sketches.	12	L2	4	1,2,3
OR					
V.	(a) Explain the layout of a diesel engine power plant with neat sketch.	6	L2	4	1,2,3
	(b) Explain the functions of following systems in diesel engine power plant. (i) Cooling system (ii) Lubricating system (iii) Fuel system	6	L2	2	1,3
VI.	(a) Explain fluidized bed combustion system with the help of a neat diagram.	6	L2	3	1,2,3
	(b) Why are multiple chimneys preferred over single unit chimneys?	6	L2	2	1,2,3
OR					
VII.	How are different types of nuclear reactor classified? Write briefly on each classification with the help of neat diagrams.	12	L2	3	1,2,3
VIII.	(a) Distinguish between thermoelectric conversion and thermionic conversion systems.	6	L2	3	1,2,3
	(b) Explain the working of a fuel cell. What are its advantages over conventional system of power generation?	6	L2	3	1,2,3
OR					
IX.	(a) Describe the working of a geothermal power plant.	6	L2	4	1,2
	(b) Explain different types of solar collectors, with neat sketches.	6	L2	4	1,2,3

Bloom's Taxonomy Levels
L1 = 20%, L2 = 70%, L3 = 10%
